

# TwinLight: A Multi-Timescale Physical-Layer-Aware T-API-Compliant Optical Domain Controller Digital Twin

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**Abstract** We demonstrate *TwinLight*, an open-source optical-network digital twin acting as an optical domain controller. Its T-API REST and telemetry streaming interfaces expose a physical-layer engine that combines a GNP steady-state baseline with four analytical transients. ©2026 The Author(s)

## Review

Disaggregated and software-defined optical networks increasingly rely on digital twins (DTs) to de-risk control-plane decisions. A faithful software replica of the optical domain helps validate software-defined networking (SDN) control logic, explore what-if scenarios, prototype routing, modulation, and spectrum assignment (RMSA) algorithms, and generate training data for machine learning (ML) models [1, 2]. This requirement becomes more relevant as autonomous and agentic strategies begin to operate in closed loop, observe physics-consistent consequences, and be validated against the same northbound interfaces that a production controller would expose [3, 4].

The tooling landscape that has emerged to meet these needs resolves into three disjoint families of solutions [5]. First, physical-layer tools such as GNP compute steady-state quality of transmission (QoT) with high fidelity [6, 7], but are not designed to expose the persistent state and standardized northbound interfaces that a controller-facing optical domain controller (ODC) must provide. Second, network emulators such as Mininet-Optical capture topology, embed optical performance monitoring (OPM), and expose NETCONF and gNMI interfaces aligned with OpenConfig and OpenROADM device models, yet operate exclusively in steady state [8]. Within the same family, data-driven digital twins such as OCATA refine steady-state QoT estimation through telemetry-driven model tuning, but expose ad-hoc northbound interfaces and likewise lack an analytical model of temporal impairments [9]. A third family combines physical-layer-aware QoT with RMSA logic and recalculates mutual impact when a new lightpath perturbs the signal-to-noise ratio (SNR) of existing services [10, 11], but lacks both a standards-compliant northbound interface and analytical models of temporal physical impairments. Considered in isolation, none of these families can serve as an ODC for a production orchestrator.

Two further developments sharpen the requirements that a controller-facing DT must satisfy. The first concerns physics, including physical effects that span multiple timescales. The community has converged on the Gaussian noise (GN) family of models as the working description of nonlinear interference in coherent transmission, and on GNP as its *de facto* implementation, with recent field validation against partially-disaggregated multi-vendor deployments [6, 12, 13]. Real coherent links, however, undergo impairments that no steady-state snapshot can capture. Channel add-drop events trigger microsecond-scale power excursions in cascaded Erbium-doped fiber amplifiers (EDFAs) [14]. The state of polarisation drifts on second-to-day timescales [15], and polarization-dependent loss (PDL) adds a stochastic optical signal-to-noise ratio (OSNR) penalty over the same range [16–18]. Accumulated dispersion couples with the local-oscillator linewidth to produce equalization-enhanced phase noise (EPPN) on the minute scale [19], while diurnal thermal cycles drift the chromatic dispersion (CD) and modulate span loss [20, 21]. Together, these mechanisms span nine orders of magnitude in time.

The second concerns the application programming interfaces (APIs) that the DT exposes. The Open Networking Foundation (ONF) Transport-API (T-API) defines a representational state transfer (REST) interface that an ODC must expose to a hierarchical orchestrator, which may integrate optical services with other technology domains such as packet-switching and wireless networks. T-API exposes topology, connectivity, photonic-media, and path-computation contexts that allow comprehensive control of the infrastructure associated with the ODC [22, 23]. T-API streaming, aligned with the OpenConfig gRPC network management interface (gNMI) subscription model, has emerged as the telemetry counterpart [24]. A DT that aims to operate under a production orchestrator must therefore implement both protocols natively.

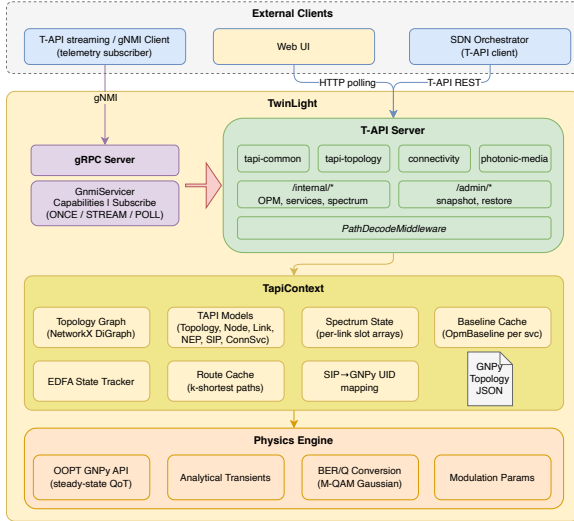


Fig. 1: TwinLight architecture.

## Novelty

The overall architecture of TwinLight is shown in Fig. 1. Northbound, the DT exposes two interfaces: (i) a T-API REST interface that can be used to interface with orchestrators and (ii) a T-API streaming interface to telemetry subscribers. The DT also provides a built-in web user interface that uses the same T-API REST surface, as well as internal endpoints for administrative tasks. A single stateful context manages the topology graph, the connectivity services, the per-link spectrum state, and the per-service QoT baselines, so that physics, state, and standard interfaces remain consistent without inter-process synchronization. The physics engine combines a GNPy steady-state baseline with the four analytical transient models and a modulation-format-aware bit-error-rate conversion, feeding back into the same context that the north-bound interfaces serve.

TwinLight's novelty lies in three compounding contributions. First, TwinLight is a DT that serves as a complete ODC, simultaneously exposing T-API and T-API telemetry streaming interfaces over a live physics engine, with both steady-state QoT and transients. The same physics engine is exposed through T-API and the built-in web user interface, such that control logic, telemetry consumers, and human operators share a single source of truth.

Second, TwinLight provides a PDL treatment appropriate for coherent receivers. The model cascades a per-hinge polarization transfer across reconfigurable optical add-drop multiplexer (ROADM) and EDFA hinges of each light-path following the hinge formulation of [16]. Each hinge is assigned a Maxwell-distributed magnitude [25] seeded deterministically from its identifier, and its alignment drifts with a slightly incommensurate period, so that the joint trajectory becomes quasi-periodic and ergodically fills the joint PDL en-

semble without an explicit Monte-Carlo pass. The accumulated PDL is then mapped to a coherent-receiver OSNR penalty using the experimentally characterized treatment of [17, 18]. The mapping is sensitive to a configurable amplifier spontaneous emission (ASE)-distribution scenario that reflects the transmitter-, distributed-, and receiver-side cases reported in the same literature.

Third, TwinLight composes four transient impairment models onto the GNPy steady-state baseline through a QoT combiner that maintains generalized signal-to-noise ratio (GSNR) and OSNR as independent metrics and routes each impairment to its physically correct counterpart. EDFA reservoir dynamics, polarisation effects, and the thermal loss component of the environmental drift act on both GSNR and OSNR. EEPN, a digital signal processing (DSP)-domain impairment, acts only on GSNR. Thermal CD drift updates the reported CD metric directly and modulates GSNR indirectly through the EEPN term, the only path by which a DSP-compensated dispersion change reaches a coherent receiver. The combiner produces a six-metric OPM tuple at every time instance, comprising OSNR, GSNR, pre-forward error correction (FEC) BER, Q-factor, CD, and polarization-mode dispersion (PMD), on top of a baseline that respects the physical separation between linear, nonlinear, polarisation, phase-noise, and environmental contributions. From the same combined state, the combiner also synthesizes a constellation diagram and an eye diagram. The diagrams are parameterized by the GSNR, the combined transmitter and local-oscillator linewidth, and the PMD, following the Gaussian-mixture representation introduced by OCATA [9, 26] and the standard PMD-driven jitter mapping [15]. The artefact thereby exposes physics-consistent observability at three abstraction levels: aggregated metrics, scatter statistics, and time-domain waveforms, all derived from the same physics state.

## ECOC Relevance

Network operators and vendors of SDN orchestrators repeatedly face the problem of integrating new control logic against a realistic, standards-compliant ODC. Such integration requires access to laboratory hardware that is expensive, scarce, and routinely committed to production-readiness validation rather than open experimentation. A free, scriptable DT that exposes the same north-bound interfaces as a real ODC removes that bottleneck and lets integration work start before any optics are powered on. Academic researchers working on RMSA, control-plane algorithms, agentic artificial intelligence (AI), ML-based QoT estimation, and failure detection and prediction gain a reproducible physical-layer baseline that can generate realistic training data on demand. The



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## References

- [1] R. Vilalta et al., "Applying Digital Twins to Optical Networks with Cloud-Native SDN Controllers", *IEEE Communications Magazine*, vol. 61, no. 12, pp. 128–134, Dec. 2023. DOI: 10.1109/MCOM.003.2300105.
- [2] D. Wang et al., "Digital Twin of Optical Networks: A Review of Recent Advances and Future Trends", *Journal of Lightwave Technology*, vol. 42, no. 12, pp. 4233–4259, Jun. 2024. DOI: 10.1109/JLT.2024.3401419.
- [3] E. Etezadi et al., "Demonstration of DRL-based intelligent spectrum management over a T-API-enabled optical network digital twin", in *European Conference on Optical Communications (ECOC)*, Oct. 2023, pp. 1730–1733. DOI: 10.1049/icp.2024.1801.
- [4] Y. Song et al., "Synergistic Interplay of Large Language Model and Digital Twin for Autonomous Optical Networks: Field Demonstrations", *IEEE Communications Magazine*, vol. 63, no. 6, pp. 90–96, Jun. 2025. DOI: 10.1109/MCOM.001.2400342.
- [5] V. V. Deshpande and S. Mohani, "A Comparative Study of Network Simulators with Focus on OTN (Optical Transport Networks)", in *IEEE Engineering Informatics*, Nov. 2023. DOI: 10.1109/IEECONF58110.2023.10520628.
- [6] A. Ferrari et al., "GNPy: An open source application for physical layer aware open optical networks", *Journal of Optical Communications and Networking*, vol. 12, no. 6, pp. C31–C40, Jun. 2020. DOI: 10.1364/JOCN.382906.
- [7] V. Curri, "GNPy model of the physical layer for open and disaggregated optical networking [Invited]", *Journal of Optical Communications and Networking*, vol. 14, no. 6, pp. C92–C104, Jun. 2022. DOI: 10.1364/JOCN.452868.
- [8] B. Lantz, A. A. Díaz-Montiel, J. Yu, C. Rios, M. Ruffini, and D. Kilper, "Demonstration of Software-Defined Packet-Optical Network Emulation with Mininet-Optical and ONOS", in *Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2020, M3Z.9. DOI: 10.1364/OFC.2020.M3Z.9.
- [9] P. Gonzalez, S. Ghasrizadeh, S. Barzegar, H. Shakespear-Miles, M. Ruiz, and L. Velasco, "Demonstrating Lightpath Operation with the OCATA Digital Twin in Multiband Optical Networks", in *International Conference on Transparent Optical Networks (ICTON)*, Jul. 2025. DOI: 10.1109/ICTON67126.2025.11125283.
- [10] C. Natalino et al., "Optical Networking Gym: An open-source toolkit for resource assignment problems in optical networks", *Journal of Optical Communications and Networking*, vol. 16, no. 12, G40–G51, Dec. 2024. DOI: 10.1364/JOCN.532850.
- [11] V. V. Garbhapu, C. Ware, and M. Lourdiane, "Physical-layer-aware network simulator for future optical functionalities", in *International Conference on Network of the Future (NoF)*, Oct. 2023, pp. 28–36. DOI: 10.1109/NoF58724.2023.10302753.
- [12] A. Carena, G. Bosco, V. Curri, Y. Jiang, P. Poggiolini, and F. Forghieri, "EGN model of non-linear fiber propagation", *Optics Express*, vol. 22, no. 13, pp. 16 335–16 362, Jun. 30, 2014. DOI: 10.1364/OE.22.016335.
- [13] G. Borracchini et al., "Experimental Demonstration of Partially Disaggregated Optical Network Control Using the Physical Layer Digital Twin", *IEEE Transactions on Network and Service Management*, vol. 20, no. 3, pp. 2343–2355, Sep. 2023. DOI: 10.1109/TNSM.2023.3288823.
- [14] A. Bononi and L. Rusch, "Doped-fiber amplifier dynamics: A system perspective", *Journal of Lightwave Technology*, vol. 16, no. 5, pp. 945–956, May 1998. DOI: 10.1109/50.669054.
- [15] J. P. Gordon and H. Kogelnik, "PMD fundamentals: Polarization mode dispersion in optical fibers", *Proceedings of the National Academy of Sciences*, vol. 97, no. 9, pp. 4541–4550, Apr. 25, 2000. DOI: 10.1073/pnas.97.9.4541.
- [16] S. Zarkosky and M. Shtaf, "Statistical distribution of polarization-dependent loss in systems characterized by the hinge model", *Optics Letters*, vol. 45, no. 5, pp. 1224–1227, Mar. 1, 2020. DOI: 10.1364/OL.385525.
- [17] A. D'Amico et al., "Experimental Probing and Modeling of the PDL Impact on the Optical Signal-to-Noise Ratio", in *Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2023, W1E.6. DOI: 10.1364/OFC.2023.W1E.6.
- [18] E. Miotto, A. D'Amico, R. Ambrosone, F. Aquilino, S. Straullu, and V. Curri, "Statistical Assessment of System Margin in Metro Networks Impaired by PDL", in *Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2025, M3F.5. DOI: 10.1364/OFC.2025.M3F.5.
- [19] W. Shieh and K.-P. Ho, "Equalization-enhanced phase noise for coherent-detection systems using electronic digital signal processing", *Optics Express*, vol. 16, no. 20, pp. 15 718–15 727, Sep. 29, 2008. DOI: 10.1364/OE.16.015718.
- [20] T. Kato, Y. Koyano, and M. Nishimura, "Temperature dependence of chromatic dispersion in various types of optical fiber", *Optics Letters*, vol. 25, no. 16, pp. 1156–1158, Aug. 15, 2000. DOI: 10.1364/OL.25.001156.
- [21] C. Berkdemir and S. Özsoy, "The temperature dependent performance analysis of EDFAs pumped at 1480 nm: A more accurate propagation equation", *Optics Express*, vol. 13, no. 13, pp. 5179–5185, Jun. 27, 2005. DOI: 10.1364/OPEX.13.005179.
- [22] C. Manso et al., "TAPI-enabled SDN control for partially disaggregated multi-domain (OLS) and multi-layer (WDM over SDM) optical networks [Invited]", *Journal of Optical Communications and Networking*, vol. 13, no. 1, A21–A33, Jan. 2021. DOI: 10.1364/JOCN.402187.
- [23] "TR-547 TAPI v2.4.0 Reference Implementation Agreement", Open Networking Foundation, TR-547, Dec. 2022.
- [24] V. Karunakaran et al., "TAPI-based Telemetry Streaming in Multi-domain Optical Transport Network", in *Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2024, M3Z.9. DOI: 10.1364/OFC.2024.M3Z.9.
- [25] A. Mecozzi and M. Shtaf, "The statistics of polarization-dependent loss in optical communication systems", *IEEE Photonics Technology Letters*, vol. 14, no. 3, pp. 313–315, Mar. 2002. DOI: 10.1109/68.986797.
- [26] D. Sequeira, M. Ruiz, N. Costa, A. Napoli, J. Pedro, and L. Velasco, "OCATA: A deep-learning-based digital twin for the optical time domain", *Journal of Optical Communications and Networking*, vol. 15, no. 2, pp. 87–97, Feb. 2023. DOI: 10.1364/JOCN.477341.